

Chapter 1

TT

1.1 A

As we did in FOL, we first need to introduce some symbols in order to construct a formal language of TT.

- ◇ $V = \{v_0, v_1, \dots\}$ is the set of term variables;
- ◇ $A = \{\alpha_0, \alpha_1, \dots\}$ is the set of type variables (below we take the union of A and V , and denote it by V);
- ◇ $C = \{c_0, c_1, \dots\}$ is the set of constants that are new definitions;
- ◇ $P = \{., :, \equiv, (,)\}$ is the set of punctuation;
- ◇ $B = \{\lambda, \Pi\}$ is the set of binders;
- ◇ $H = \{*, \square\}$ is the set of higher order types.

Then we let

$$F = V|H|FF|BV : F.F|CF^*$$

in which F^* is the set of words over set F , including the empty word $_$.

Definition 1.1.1 A definition D has the form

$$\mathbf{x} : \mathbf{A} \quad c\mathbf{x} \equiv d(\mathbf{x}) : a$$

where we denote $\mathbf{x}$ by $(x_i)_{i=0}^s$ and $\mathbf{x} : \mathbf{A}$ by

$$x_0 : A_0 \quad x_1 : A_1 \quad \dots \quad x_s : A_s$$

note that $d(\mathbf{x})$ refers $d \in F$ and $\text{Fr } d \subset \mathbf{x}$, that $x_i \in V, c \in C$ and that all A_i, d, a are in F .

A environment Δ has the form $D_0 D_1 \dots D_n$.

Definition 1.1.2 Given $\Delta = D_0 D_1 \dots D_n$, $R_\Delta \subset F^2$ is the smallest relation satisfies the following properties:

- ◇ contains $\{(c\mathbf{U}, d(\mathbf{U})) : \mathbf{x} : \mathbf{A} \quad c\mathbf{x} \equiv d(\mathbf{x}) : a \in D \in \Delta \text{ and } \mathbf{U} \in F^*\}$;
- ◇ closed under the following operations: $F-, -F, \lambda V : F. -, Cp_1 - p_2$.

Definition 1.1.3 R_β is the smallest relation satisfies

- ◇ contains $\{((\lambda x : a. \varphi)\chi, \varphi_x^X) : x \in V \text{ and all } \varphi, \chi, a \in F\}$;
- ◇ closed under the following operations: $F-, -F, BV : F. -, BV : -. F, Cp_1 - p_2$.

Definition 1.1.4 $\sim_{\Delta, \beta} \subset F^2$ is the equivalence generated by $R_\Delta \cup R_\beta$.

Definition 1.1.5 A judgment has the form

$$\Delta \quad \mathbf{x} : \mathbf{A} \quad \varphi : a$$

All legal judgments are derived through a series of rules, as listed in follows. By the way, the question that whether a judgment is legal is decidable. Below we let $s \in H$ if it is not specified.

(Sort)

$$_ _ _ * : \square$$

(Var)

If $x \in V \setminus \mathbf{x}$, then

$$\begin{array}{l} \Delta \quad \mathbf{x} : \mathbf{A} \quad a : s \\ \Delta \quad \mathbf{x} : \mathbf{A} \quad x : a \quad x : a \end{array}$$

(Weak)

If $x \in V \setminus \mathbf{x}$, then

$$\begin{array}{l} \Delta \quad \mathbf{x} : \mathbf{A} \quad \varphi : a \\ \Delta \quad \mathbf{x} : \mathbf{A} \quad b : s \\ \Delta \quad \mathbf{x} : \mathbf{A} \quad x : b \quad \varphi : a \end{array}$$

(Form)

$$\begin{array}{l} \Delta \quad \mathbf{x} : \mathbf{A} \quad a_1 : s_1 \\ \Delta \quad \mathbf{x} : \mathbf{A} \quad x : a_1 \quad a_2 : s_2 \\ \Delta \quad \mathbf{x} : \mathbf{A} \quad \Pi x : a_1.a_2 : s_2 \end{array}$$

(Appl)

$$\begin{array}{l} \Delta \quad \mathbf{x} : \mathbf{A} \quad \varphi : \Pi x : a.b \\ \Delta \quad \mathbf{x} : \mathbf{A} \quad \chi : a \\ \Delta \quad \mathbf{x} : \mathbf{A} \quad \varphi \chi : b_x^\chi \end{array}$$

(Abst)

$$\begin{array}{l} \Delta \quad \mathbf{x} : \mathbf{A} \quad x : a \quad \varphi : b \\ \Delta \quad \mathbf{x} : \mathbf{A} \quad \Pi x : a.b : s \\ \Delta \quad \mathbf{x} : \mathbf{A} \quad (\lambda x : a.\varphi) : (\Pi x : a.b) \end{array}$$

(Def)

$$\begin{array}{l} \Delta \quad \mathbf{x} : \mathbf{A} \quad \varphi : a \\ \Delta \quad \mathbf{y} : \mathbf{B} \quad d(\mathbf{y}) : b \\ \Delta D \quad \mathbf{x} : \mathbf{A} \quad \varphi : a \end{array}$$

where D is the string

$$\mathbf{y} : \mathbf{B} \quad c\mathbf{y} \equiv d(\mathbf{y}) : b$$

that $d(\mathbf{y})$ could be an empty word.

(Inst)

If Δ contains the element

$$\mathbf{y} : B \quad c\mathbf{y} \equiv d(\mathbf{y}) : a$$

that $d(\mathbf{y})$ could be empty, and we have $U \in F^*$ with the same size of $\mathbf{y}$, then

$$\begin{array}{l} \Delta \quad \mathbf{x} : A \quad * : \square \\ \Delta \quad \mathbf{x} : A \quad U_0 : B_0 \frac{U}{\mathbf{y}} \\ \vdots \\ \Delta \quad \mathbf{x} : A \quad U_m : B_m \frac{U}{\mathbf{y}} \\ \Delta \quad \mathbf{x} : A \quad cU : a \frac{U}{\mathbf{y}} \end{array}$$

(Conv)

If $a \sim_{\Delta, \beta} a'$, then

$$\begin{array}{l} \Delta \quad \mathbf{x} : A \quad \varphi : a \\ \Delta \quad \mathbf{x} : A \quad a' : s \\ \Delta \quad \mathbf{x} : A \quad \varphi : a' \end{array}$$

(Derived: Par)

Given Δ and let D be

$$\mathbf{y} : B \quad c\mathbf{y} \equiv d(\mathbf{y}) : a$$

satisfies that c does not occur in Δ , then

$$\begin{array}{l} \Delta \quad \mathbf{x} : A \quad d(\mathbf{y}) : a \\ \Delta D \quad \mathbf{x} : A \quad c\mathbf{y} : a \end{array}$$